ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. (CSE)

Semester: II

SUBJECT CODE: BME202

BASIC CIVIL & MECHANICAL ENGINEERING

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. **(10 Marks)**

Q. 1. Define the following questions: **(1 × 5 = 5 Marks)**

(i) Define normal stress and give an example.

(ii) What is the difference between shear stress and normal stress?

(iii) State Hooke's Law and its significance in material mechanics.

(iv) What is Poisson's ratio? Provide its formula.

(v) What is casting process?

Q. 2. Answer the following questions: **(1 × 5 = 5 Marks)**

(i) Define engineering materials.

(ii) What is the difference between metallic and non-metallic materials?

(iii) What are ferrous materials? Give an example.

(iv) What is the significance of ductility in materials?

(v) Define a thermodynamic system with an example.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Discuss the First Law and second law of Thermodynamics in detail. How is it applied in real-world thermodynamic system?

OR

(b) What are the merits and demerits of two-stroke engines compared to four-stroke engines?

(ii)(a) Explain the concept of positive, negative, and zero processes in manufacturing.

OR

(b) Discuss the significance of the elastic constants in material mechanics and relation between them.

Q.4.(i)(a) Discuss the types of **beams** and the various **loads** they are subjected to. Provide examples for each.

OR

(b) What is force and force system? Explain in detail.

(ii)(a) A steel wire of length 2 m and diameter 1 mm is stretched by a force of 100 N. The elongation produced in the wire is measured to be 0.1 mm. Calculate the Young's Modulus (E), Poisson's Ratio (ν), and the Bulk Modulus (K) for the material of the wire. Assume Poisson's ratio is assumed to be 0.3 for steel.

OR

(b) What is the difference between heat engine, refrigerator, and heat pump?

SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) Explain the construction and working of a two-stroke SI engine. Compare it with the four-stroke SI engine in terms of performance and efficiency.

OR

(b) Discuss the concept of hybrid vehicles. How do they combine internal combustion engines and electric motors?

Q.6.(a) Describe the working principle of a refrigerator. Discuss its main components and the concept of Co-efficient of performance (COP).

OR

(b) Discuss the different types of engineering materials and their properties. How does material selection affect the manufacturing process?